An **AI Outbound Orchestrator** that:

takes a website → infers ICP → finds leads via Apollo/Ampleleads → verifies emails via MillionVerifier → creates campaigns in Instantly → writes personalized sequences using LLMs → runs sending + LinkedIn touches → updates status & notifies owner.

1. High-level architecture

Components (logical):

- **Frontend / Input**
  - submit website URL, set campaign parameters (region, max leads, cadence), connect provider accounts (Apollo, Ampleleads, MillionVerifier, Instantly, LinkedIn accounts), configure inbox pool.
- **Orchestrator / Workflow Engine**
  - n8n flows coordinate the stages and call microservices.
- **AI Brain**
  - AI agents that perform website analysis, ICP inference, personalization/copywriting, and decision logic.
- **Integrations**
  - Connectors for Apollo, Ampleleads, MillionVerifier, Instantly, LinkedIn (via official APIs), Google Sheets / cloud storage.
- **Storage & DB**
  - Relational DB + object storage for CSVs, job logs, lead snapshots.
- **Campaign Manager**
  - Logic to allocate inboxes, schedule sends, track quotas & deliverability.
- **Monitor & Alerting**
  - Dashboards, logs, and on-call alerts.
- **Auth & Secrets**

- OAuth where available; secrets manager.

➔ **Notification**

- Email / Slack / webhook to notify the owner of “interested” replies.

Flow diagram (textual):

1. **Input:** Website URL + campaign params → create Job.
2. **Website Scraper** → structured company profile.
3. **ICP Engine (AI)** → target roles, company size, geography, pain points.
4. **Build Apollo search query** → call Apollo API → produce search link.
5. Optionally pass search link to Amplemarket to generate lead batches (async 24–48h).
6. **When leads ready** → fetch as spreadsheet → normalize fields → store.
7. **Email Verification:** send CSV to MillionVerifier → receive verdicts → keep valid.
8. **Write copy:** for each lead, AI reads site + LinkedIn + socials → produces personalized subject, first line, body, follow-ups.
9. **Campaign creation:** create campaign in Instantly via API → upload verified leads.
10. **Send scheduling & LinkedIn touch automation** (3 hours after email → connection request).
11. **Monitor replies** → classify (OOO/interested/not interested) → update DB → alert owner when interested.

2. Concrete requirements

Functional requirements

➔ Accept website URL and campaign configuration.
➔ Auto-detect ICP and generate a search filter for Apollo.
➔ Pull leads from Apollo / Ampleleads (support both).
➔ Normalize and store lead data in CSV.
➔ Verify emails using MillionVerifier and filter Valid.
➔ Auto-generate personalized email sequences using LLMs per lead (subject, first line, body, 2–4 follow-ups).
➔ Create & schedule campaigns in Instantly and assign inboxes.
➔ LinkedIn touch: send connection request 2 hours after first email.
➔ Monitor replies & auto-classify (OOO / interested / not interested / spam / other).
➔ Alert owner by email when “interested”.
➔ Audit log for every decision & generated text (for review).

Non-functional requirements

- **Scalable:** handle X leads/day (configurable).
- **Secure:** encrypted secrets, GDPR / CAN-SPAM compliant opt-outs.
- **Reliable:** retry & backoff for API failures; transactional consistency between steps.
- **Observable:** durable job logs, dashboards, SLA alerts.
- **Extensible:** plugin drivers for new providers.

External integrations (minimum)

- Apollo API (lead search)
- Ample Leads API (lead fulfillment)
- MillionVerifier (email verification)
- Instantly.ai (campaign creation & reply detection)
- LinkedIn (automation via API or safe headless flows)
- LLM provider (OpenAI/ Anthropic / local LLM) via LangChain
- n8n as the orchestrator (or custom scheduler)

3. Data model & file formats

Lead CSV columns

- lead_id (string)
- first_name
- last_name
- title
- company_name
- company_website
- company_linkedin
- person_linkedin
- email
- email_verification_status (valid / risky / invalid)
- city
- country
- industry
- company_size_bucket (1-10,11-50,51-200,201-1000,1000+)
- source (apollo / ampleleads)
- search_link (original search URL)
- added_at (ISO8601)
- notes (text) - used by AI to store the “personalized first line” excerpt
- status (pending / uploaded / sent / replied / bounced / unsubscribed)

- ➔ reply_label (interested / not_interested / OOO / spam / follow_up_required)
- ➔ ai_generated_subject
- ➔ ai_generated_firstline
- ➔ ai_generated_body
- ➔ ai_generated_followups (JSON array)

Campaign object (DB)

- ➔ campaign_id
- ➔ owner_id
- ➔ website_input
- ➔ icp_definition (JSON)
- ➔ lead_count
- ➔ inbox_pool (list of email accounts)
- ➔ cadence_config (days between follow-ups)
- ➔ send_limits_per_inbox_per_day
- ➔ status
- ➔ created_at, updated_at

4. LLM prompts & prompt engineering

ICP inference (system prompt outline)

```
None
System: You are an ICP detection assistant. Input: company
website HTML/text. Output: JSON with industry, primary
product/service, company_size_bucket, target_geographies, likely
buyer_roles (array), pain_points (array of 3), ideal messaging
angle (1-2 lines).
```

Personalized first line template

```
None
Input: lead record (company + person + website + linkedIn latest
post)
```

```
Task: produce a factual first line (<=28 words) referencing a
specific detail (e.g., "Congrats on the Series A", or "I liked
your latest post on X").
Tone: professional, concise, curiosity-sparking.
Output: single sentence.
```

Email generation (single lead)

```
None
System: You are a top B2B copywriter for [ICP]. Write a 120–160
word cold outreach email that includes:
- a 4–7 word subject line,
- 1-sentence personalized first line referencing a fact
(provided),
- 2-line value prop tailored to pain points,
- 1-line social proof,
- one clear CTA (book a 15-min call / quick intro).
Return JSON: {subject, body, first_line, cta_text}
```

Follow-up rewrite (triggered when new activity detected)

```
None
Input: previous message history, new signal (job change / funding
/ product launch)
Task: produce single follow-up paragraph referencing the new
signal and restating CTA.
```

Store all prompts in version control. Add a safety guardrail prompt to forbid fabricating claims about the prospect (e.g., "do not invent awards/funding").

5. n8n Flow Blueprint)

Nodes (sequence):

1. **HTTP Trigger** - receive website + config
2. **Website Scraper Node** - fetch homepage, run HTML→text (or call a microservice)
3. **LLM ICP Node** - call LangChain endpoint → returns ICP JSON
4. **Build Apollo Query Node** - use ICP to build field filters (role, seniority, company size)
5. **Apollo API Node** - create search + save search_link
6. **Amplemarket / Lead Fulfillment Node** - send search_link (async)
7. **Poll Node** - every 6 hours check if leads ready
8. **Fetch Leads Node** - get CSV → upload to storage + normalize
9. **MillionVerifier Node** - send CSV → poll result
10. **Filter Node** - only valid emails → save verified CSV
11. **LLM Personalization Batch Node** - for chunked leads (e.g., 200 per batch) call AI to create personalized email+followups
12. **Instantly API Node** - create campaign → upload leads with personalization as variables
13. **Schedule Node** - calculate send windows based on timezone & inbox allocation
14. **LinkedIn Touch Node** - queue connection request (delayed 3 hours)
15. **Monitor Replies Node** - webhook handling from Instantly for replies; call LLM to classify reply; update DB + send notification if interested

Keep all steps idempotent and log job ids from external APIs.

6. Inbox allocation algorithm

Inputs: total_leads N, daily_send_limit_per_inbox L, desired_days_to_send D.

1. required_daily_sends = ceil(N / D)
2. required_inboxes = ceil(required_daily_sends / L)
3. Add a safety buffer 20% to required_inboxes.
4. Assign inboxes prioritizing reputation (older inboxes first).
5. Track per-inbox warmup stage: if new inbox, limit first week sends to 10/day and ramp.

Implement a state machine per inbox: warmup → normal → throttled → paused.

7. Error cases, detection & recovery

A. Website analysis fails (bad HTML / JS heavy)

- ➔ **Symptoms**: empty or gibberish ICP.
- ➔ **Detection**: LLM confidence low; missing required keys.
- ➔ **Recovery**:
    - ◆ Retry with headless browser (render JS).
    - ◆ Fallback: ask users to manually pick ICP from suggested options.
    - ◆ Log and notify users.

B. Apollo API failures / rate limit

- ➔ **Symptoms**: 429, 5xx, timeouts.
- ➔ **Recovery**:
    - ◆ Exponential backoff + retry up to N times.
    - ◆ If repeated, switch to an alternate data source (Amplemarket or Clay).
    - ◆ Notify operator if > X minutes delay.

C. Ampleleads lead gen takes >48h or fails

- ➔ **Symptoms**: no leads after expected time.
- ➔ **Recovery**:
    - ◆ Poll with backoff.
    - ◆ If still missing, fetch a partial list from Apollo export instead.
    - ◆ Notify users and offer to continue with a partial list.

D. Verification service partial results / high risky rate

- ➔ **Symptoms**: >30% risky/unknown.
- ➔ **Recovery**:
    - ◆ Re-run verification using alternate providers for problematic addresses.
    - ◆ If still high, reduce campaign volume and pause until the issue is analyzed.

E. Instantly campaign creation fails / invalid CSV

- ➔ **Symptoms**: 400 errors, missing required fields.
- ➔ **Recovery**:

- Validate CSV schema locally before upload.
- If API fails, retry and upload via chunked batches.
- Maintain transaction log so duplicates don't re-send.

F. Deliverability / high bounce or spam complaints

- **Symptoms**: >5% bounce, >0.1% spam complaints in a short window.
- **Recovery**:

  - Pause campaign automatically.
  - Quarantine the inbox responsible, start deliverability checks (SPF/DKIM/DMARC).
  - Reduce sending volume and warm up new inboxes gradually.
  - Rotate to fresh inboxes only after warmup and manual validation.

G. LinkedIn automation throttled / account restricted

- **Symptoms**: action blocked, account warnings.
- **Recovery**:

  - Back off LinkedIn automation immediately.
  - Wait for account cooldown, reduce daily actions, introduce human-like delays.
  - Prefer organic connection sequences (manual step) for risky accounts.

H. LLM hallucination (fabricated facts)

- **Symptoms**: AI produces incorrect first line claiming nonexistent events.
- **Detection**: QA rules: must match one of scraped facts or LinkedIn text snippet.
- **Recovery**:

  - Add verification step: every generated personalized first line must include a direct quote snippet from source (website or LinkedIn). If not, mark as generic non-specific safe first line.
  - Keep audit trail: source_url + snippet for each first line.

I. Reply classification mistakes

- **Symptoms**: false positives/negatives on "interested".
- **Recovery**:

  - Human-in-the-loop review for any "interested" before auto-notifying high-value owners (or require owner confirmation).
  - Retrain classifier with labelled replies.

8. Security, privacy & compliance

- **Secrets**: store API keys in Vault / cloud KMS; rotate monthly.
- **Authentication**: OAuth2 for providers (Apollo, Instantly) where available.
- **Data encryption**: at rest (AES-256) and in transit (TLS1.3).
- **PII minimization**: only store required lead fields; retention policy (e.g., delete raw CSVs after 90 days unless user opts in).
- **Opt-out handling**: honor unsubscribe links and maintain suppression lists across all campaigns.
- **GDPR**: if EU prospects, ensure lawful basis (legitimate interest) and allow data removal requests; keep processing records.
- **Audit trail**: store every LLM prompt/response + source snippet for legal traceability.
- **LinkedIn rules**: document risk of automation; include manual fallback flows.

9. Observability & SLOs

- **Metrics to track**
  - Leads ingested per day
  - Verified % (valid emails / total)
  - Send volume per inbox per day
  - Bounce rate, complaint rate
  - Reply rate, conversion rate (interested / booked)
  - LLM failure/hallucination rate
  - API latency & error rates per provider
- **Alerts**
  - Bounce > 5% → critical
  - Job stuck > 24 hours → warning
  - Instantly reply classification > 90% unknown → alert
- **Dashboards**
  - Campaign health (status, leads, sends, replies)
  - Deliverability & inbox reputation
  - LLM usage & cost

10. Testing strategy

- ➔ **Unit tests**: for all data transforms & schema validators.
- ➔ **Integration tests**: run on sandbox accounts for Apollo, MillionVerifier, Instantly.
- ➔ **E2E tests**: full pipeline with synthetic leads (test inboxes).
- ➔ **Fuzz tests**: malformed CSV / unexpected website inputs.
- ➔ **LLM prompt tests**: automated tests asserting required JSON keys & safety checks.
- ➔ **Manual QA**: human review of 5% of emails for the first 14 days.

11. Deployment

- ➔ **Containerization**: Docker for microservices.
- ➔ **Orchestration**: Kubernetes (GKE/EKS) for scaling workers (n8n can be self-hosted).
- ➔ **DB**: Postgres managed (e.g., Cloud SQL).
- ➔ **Object storage**: S3-compatible.
- ➔ **Secrets**: Vault / AWS Secrets Manager.
- ➔ **CI/CD**: GitHub Actions → build → test → deploy to staging & production.
- ➔ **Backups**: nightly DB backups + retention 30–90 days.

12. Cost estimate

(assume small production: 2k leads/month)

- ➔ LLM usage (OpenAI/Anthropic): $200–$1,500 depending on model & prompt volume
- ➔ Apollo / Amplemarket credits: $200–$1,000 (depends on plan)
- ➔ MillionVerifier: $50–$300 (per verification pricing)
- ➔ Instantly.ai: $50–$300 (platform & seats)
- ➔ Hosting & DB: $100–$600
- ➔ Monitoring & logging: $50–$300
- ➔ Misc (LinkedIn proxies, headless browsers): $50–$200

  Total (small): ~$700–3,200 / month

Scale linearly or better with optimizations (batching, caching).

13. MVP roadmap

Phase 0 - Discovery & infra

- ➔ Specs, accounts, sandbox keys, basic UI
- ➔ Simple n8n pipeline to fetch leads & verify

Phase 1 - MVP

- ➔ Website → ICP inference (LLM)
- ➔ Apollo search + fetch leads
- ➔ MillionVerifier integration + filtered CSV
- ➔ LLM personalized email generation (single template)
- ➔ Instantly campaign creation & upload
- ➔ Notifications for “interested” (manual threshold)
- ➔ Basic logging & metrics

Phase 2 - Improve personalization & safety

- ➔ LinkedIn scraping + first-line verifier
- ➔ Inbox allocation & warmup logic
- ➔ Follow-up dynamic rewrite
- ➔ Reply classification + HITL

Phase 3 - Scale & polish

- ➔ Multi-agent orchestration (CrewAI)
- ➔ Advanced deliverability monitoring, automatic remediation
- ➔ Billing, multi-tenant support, full compliance

Website URL
↓
AI analyzes website → finds ICP
↓
Search leads in Apollo
↓
Export leads
↓
Verify emails
↓
Create campaign in Instantly
↓

Generate personalized email
↓
Upload leads to campaign

14. Operational playbook

- ➔ **Daily:** Review campaigns dashboard, alerts, and top replies.
- ➔ **Weekly:** Inspect deliverability metrics, rotate inboxes as needed.
- ➔ **Monthly:** Rotate API keys, review LLM prompt performance, retrain reply classifier.
- ➔ **Incident:** On bounce/complaint spike → pause affected campaigns → run deliverability checklist (SPF/DKIM/DMARC, domain warmup history, content changes).

15. Example API call & pseudocode

(1) Build Apollo query - pseudo

```python
# inputs: icp = {"roles":["Head of Growth"],
"industry":"ecommerce", "company_size":"51-200",
"countries":["UK"]}
apollo_query = {
  "filters": {
    "roles": icp["roles"],
    "industry": icp["industry"],
    "employee_count": icp["company_size"],
    "location": icp["countries"]
  },
  "limit": 1000
}
resp = apollo.create_search(apollo_query)  # store search_id
```

(2) Verify emails with MillionVerifier (pseudo)

```python
upload = million.upload_csv("leads.csv")
job = million.start_verification(upload.id)
# poll until job.completed
results = million.get_results(job.id)
valid_emails = [r for r in results if r.status == "valid"]
```

(3) Create Instantly campaign

```python
campaign = instantly.create_campaign(name="campaign-x",
from_pool=inbox_pool)
instantly.upload_contacts(campaign.id, verified_csv,
personalization_vars=True)
instantly.start_campaign(campaign.id)
```

(4) Personalize email (LangChain agent)

- ➔ Pass lead + scraped snippets to agent
- ➔ Agent returns JSON with subject / first_line / body / followups
- ➔ Validate first_line contains a source snippet (or fallback to generic safe line)

16. Human guards & safety

- ➔ Require human approval for first deployments to high-value lists.
- ➔ Provide UI to review & edit a sample of generated sequences before sending (e.g., first 50 leads).
- ➔ Keep manual pause/resume controls.
- ➔ Logging of every AI-generated line + direct source link for audit.

17. KPIs to measure success

- ➔ Leads → verified% (target > 60%)
- ➔ Open rate (target depends on industry; >20% initial)
- ➔ Reply rate (target > 2–5% for cold)
- ➔ Conversion to qualified lead (target > 0.5–2%)
- ➔ Cost / qualified lead

18. Common pitfalls & how to avoid them

- ➔ **Overpersonalization that fabricates facts** - enforce snippet verification.
- ➔ **Too many emails / fast ramping** - implement inbox warmup & limits.
- ➔ **LinkedIn blocks** - use conservative action rates & human-like delays.
- ➔ **Single provider dependency** - implement provider fallbacks (e.g., Apollo → Amplemarket → Clay).
- ➔ **No audit trail** - store prompt+response and source snapshot for each generated line.

19. Deliverables to engineers

1. This document + glossary.
2. n8n flow JSON blueprint (create nodes per section 6).
3. Postgres DB schema for Campaign & Lead objects (DDL).
4. Sample CSV schema + validator (code snippet).
5. LangChain agent prompt files (versioned).
6. CI pipeline checklist + staging credentials.
7. Monitoring dashboard wireframe (Grafana panels).
8. Testing matrix (unit / integration / e2e cases).

20. Quick checklist

- ➔ Create sandbox accounts & API keys (Apollo, Ampleleads, MillionVerifier, Instantly)
- ➔ Infra: Postgres, object storage, secrets manager
- ➔ Deploy n8n and a LangChain service
- ➔ Implement website scraper + ICP prompt
- ➔ Apollo search + fetch leads + normalize
- ➔ MillionVerifier integration + verified filtering
- ➔ LLM email generation + snippet verification guard
- ➔ Instantly campaign creation + upload
- ➔ Simple inbox allocation + warmup rules
- ➔ Reply webhook + basic classifier + email notification
- ➔ Logging, error retries, and simple dashboards
- ➔ Legal: privacy policy + suppression list + opt-outs

21. Final recommendations & next step

- ➔ Start with the **smallest workable MVP**: website → ICP → Apollo → verify → Instantly upload + one templated LLM email. Get that working end-to-end with human review. Once stable, incrementally add the per-lead personalization, LinkedIn touches, auto classification and inbox allocation.
- ➔ Put **safety nets** around every LLM-generated personalization (source verification) and deliverability logic (automatic pause).
- ➔ Track the metrics above and iterate on prompts and warmup parameters based on early results.

End of Document